



PERSONAL PROJECT · .NET 10 + LOCAL AI

LearnEasy

A friendly, AI-powered spelling tutor for kids — that runs fully offline.

WPF · Ollama (phi4-mini) · Piper neural TTS · PostgreSQL · .NET Aspire

WHY I BUILT IT

A patient teacher — without the cloud

- Kids learn spelling best by **ear and by doing** — hear the word, try it, get a gentle nudge, try again.
- Most apps need **constant internet, accounts or subscriptions** to do that.
- I wanted a **calm, private, offline buddy** a child can use at the kitchen table.



*“Can you spell...
rabbit?”*

FEATURES

What it does



Speaks every word

Neural text-to-speech reads each word aloud.



Progressive hints

Letter count → first letter → syllables → sounds.



Adapts difficulty

Moves up or down based on recent answers.



Points & badges

Streaks, levels and rewards keep kids motivated.



Tracks progress

Every attempt & score is stored in PostgreSQL.

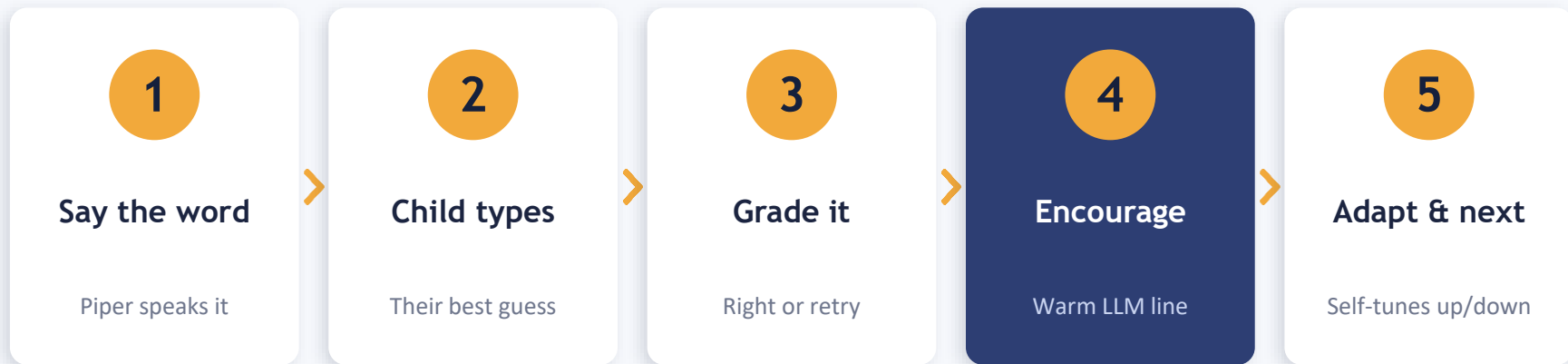


Kid-friendly UI

Big, playful, distraction-free WPF interface.

HOW IT WORKS

It behaves like a teacher



Hints and encouragement come from a local LLM — every AI step has a deterministic fallback, so a lesson never stalls.

ARCHITECTURE

Clean, modular, testable

LearnEasy.Wpf — Kid-friendly UI (MVVM)

speak · type · feedback

▼ depends on ▼

LearnEasy.Services — The “teacher”

lesson loop · adaptive · gamification · TTS · NLP

LearnEasy.Core

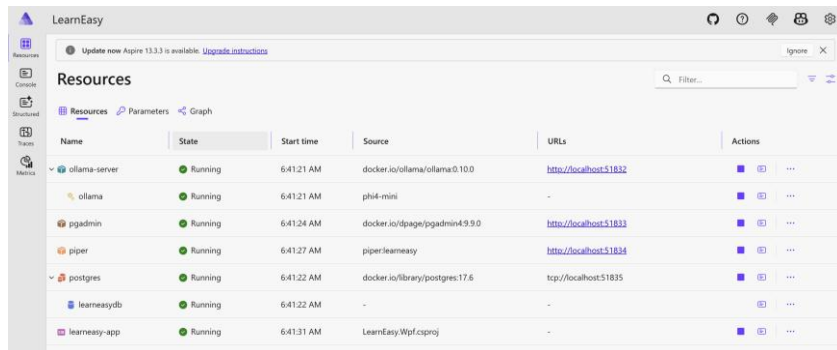
domain models + interfaces

LearnEasy.Data

EF Core + PostgreSQL

.NET Aspire orchestrates PostgreSQL · Ollama (phi4-mini) · Piper TTS (Docker)

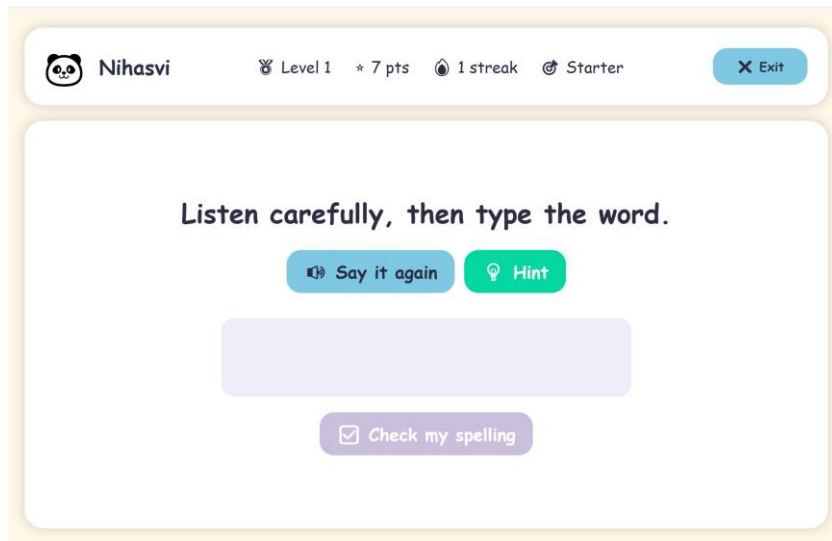
Aspire Dashboard



The screenshot shows the Aspire Dashboard interface. At the top, there's a notification bar indicating an update for Aspire 13.3.3. Below this, the 'Resources' tab is selected, showing a list of running services. The table includes columns for Name, State, Start time, Source, URLs, and Actions. The resources listed are ollama-server, ollama, pgadmin, piper, postgres, leaneasydb, and leaneasy-app, all of which are in a 'Running' state.

Name	State	Start time	Source	URLs	Actions
ollama-server	Running	6:41:21 AM	docker.io/ollama/ollama:0.10.0	http://localhost:51832	[Stop] [Refresh] [More]
ollama	Running	6:41:21 AM	phi4-mini	-	[Stop] [Refresh] [More]
pgadmin	Running	6:41:24 AM	docker.io/page/pgadmin4:9.9.0	http://localhost:51833	[Stop] [Refresh] [More]
piper	Running	6:41:27 AM	piper/leaneasy	http://localhost:51834	[Stop] [Refresh] [More]
postgres	Running	6:41:22 AM	docker.io/library/postgres:17.6	tcp://localhost:51835	[Stop] [Refresh] [More]
leaneasydb	Running	6:41:22 AM	-	-	[Stop] [Refresh] [More]
leaneasy-app	Running	6:41:31 AM	LearnEasy.Wpf.csproj	-	[Stop] [Refresh] [More]

Learn Easy Application



The screenshot shows the 'Learn Easy Application' interface. At the top, the user's profile 'Nihasvi' is displayed along with their level (Level 1), points (7 pts), streak (1 streak), and starter status. An 'Exit' button is in the top right. The main content area features the instruction 'Listen carefully, then type the word.' Below this are two buttons: 'Say it again' with a speaker icon and 'Hint' with a lightbulb icon. A large, empty text input field is provided for the user to type the word. At the bottom, there is a 'Check my spelling' button with a checkmark icon.

Nihasvi Level 1 ★ 7 pts 💧 1 streak 🌀 Starter [Exit](#)

Listen carefully, then type the word.

[Say it again](#) [Hint](#)

[Check my spelling](#)

TECH STACK

Built with



.NET 10 · WPF (MVVM)

Native Windows desktop UI



PostgreSQL · EF Core

Words, attempts, scores & badges



Ollama · phi4-mini

Local NLP — hints & encouragement



piper1-gpl

Open-source neural TTS, containerized



.NET Aspire

One-command orchestration & dev loop

DESIGN PRINCIPLE

Offline-first & resilient



100%

local

LLM + TTS run on-device. No cloud, no data leaves the machine.



0

blocked lessons

Every AI step has a rule-based fallback if a model is cold.



1

command

dotnet run spins up the whole stack via .NET Aspire.

WHAT'S NEXT

Where it's heading

- Multiplayer spelling bee mode
- Handwriting recognition (write, don't type)
- More voices & languages via swappable TTS



Designed & built end-to-end with Claude Code.

Happy to talk architecture, local LLMs or .NET Aspire — comment or DM.